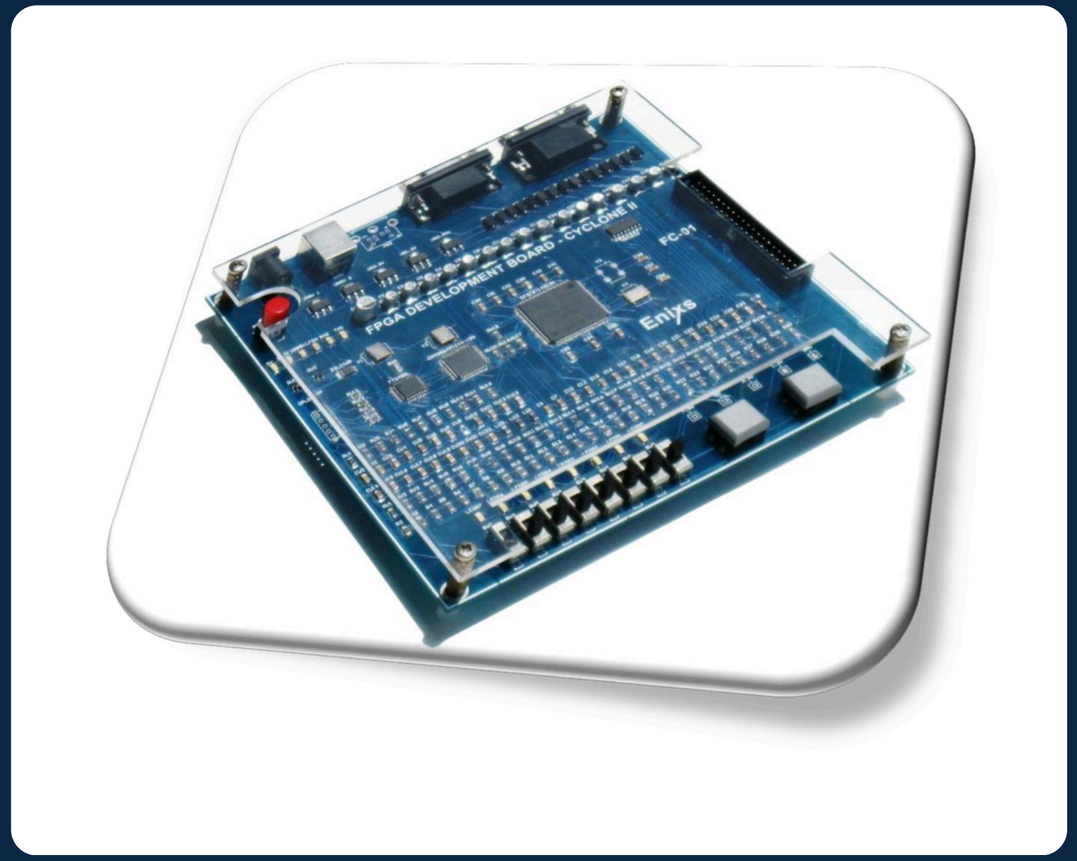
Token-Based Service Centre Controller

EC3552 – VLSI Design Mini Project

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Enixs FC-01 · Cyclone II EP2C8T144C8

Problem and solution

The problem

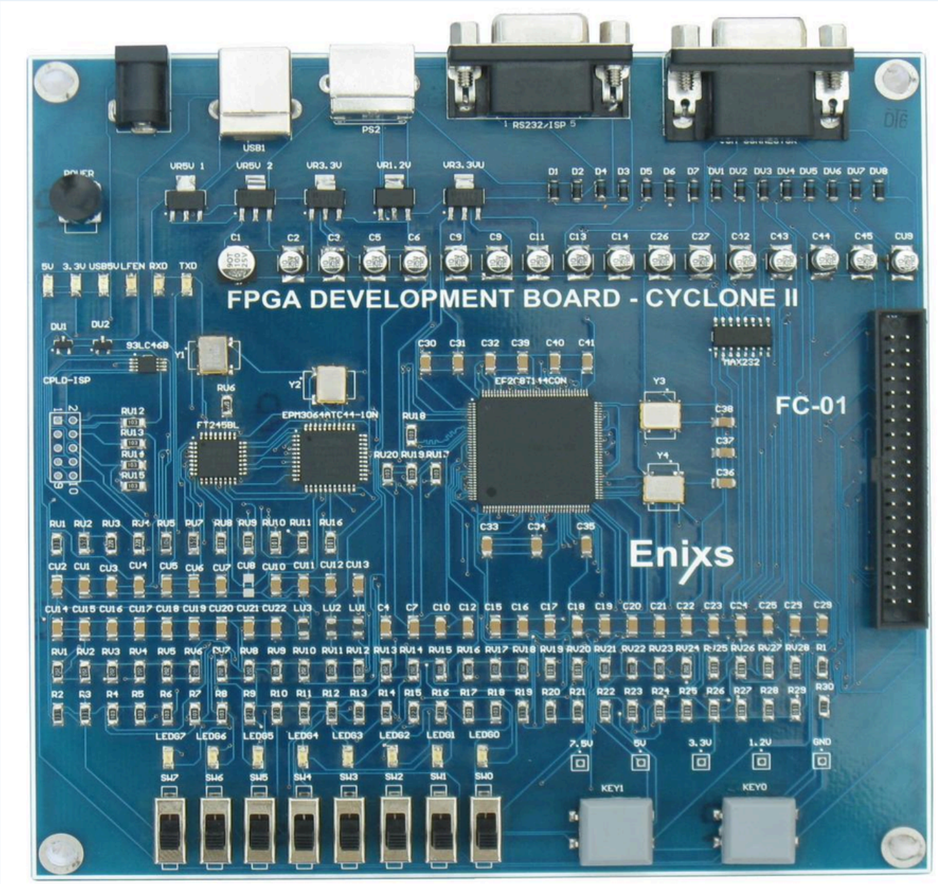
- ✗ Slow
- ✗ Difficult to manage
- ✗ Prone to confusion
- ✗ Difficult to track

The solution

- ✓ Sequential token generation
- ✓ Operator calls the next customer
- ✓ Current token shown on LEDs
- ✓ Reset and service-mode control

Manual queue → **Digital token generation** → **FPGA-based controller**

FPGA platform



Cyclone II EP2C8T144C8

8,256 logic elements	**36** M4K memory blocks	**165,888** RAM bits
18 multipliers	**2** PLLs	**144** pins

Used by this design

50 MHz clock · KEY0 · SW0 / SW1 · LEDG0–LEDG7

RTL signal pipeline

KEY0 → synchronise → debounce → detect edge → count → LEDG0–LEDG7

1

Synchronizer

Protects the clock domain from the asynchronous button input.

2

Debouncer

10 ms filter: one physical press gives one digital event.

3

Edge detector

Turns the stable press into a single next-token pulse.

4

BCD counter

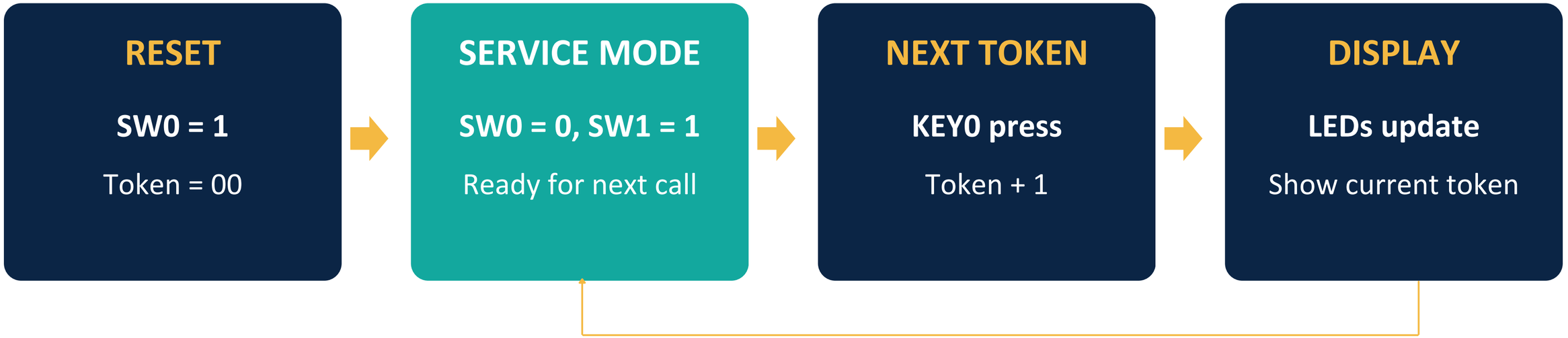
Counts 00 to 99 as human-friendly decimal tokens.

5

LED driver

Shows the two BCD digits on the board's 8 LEDs.

Token service sequence



Token sequence (wraps after 99)

00	01	02	03	…	97	98	99	00

FPGA implementation

Quartus II 13.0.1 · EP2C8T144C8 · compilation successful

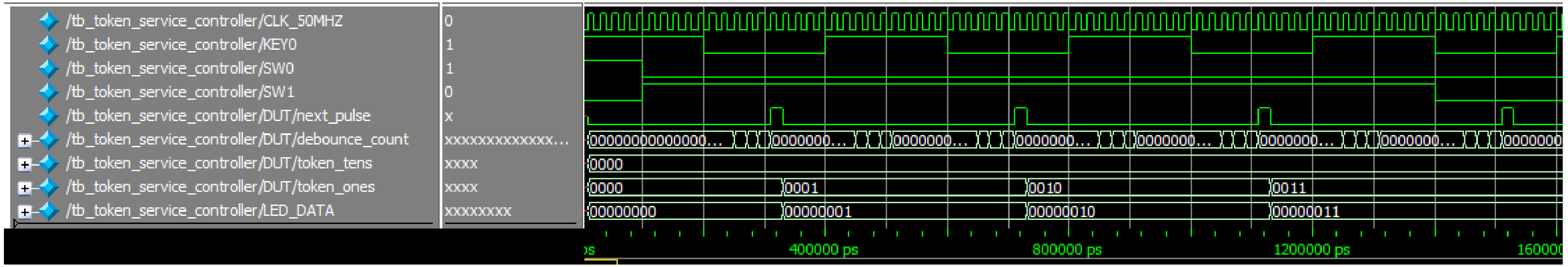
Flow Summary	
Flow Status	Successful - Fri Oct 09 08:34:11 2026
Quartus II 64-Bit Version	13.0.1 Build 232 06/12/2013 SP 1 SJ Web Edition
Revision Name	token_service_controller
Top-level Entity Name	token_service_controller
Family	Cyclone II
Device	EP2C8T144C8
Timing Models	Final
Total logic elements	46 / 8,256 (< 1 %)
Total combinational functions	46 / 8,256 (< 1 %)
Dedicated logic registers	31 / 8,256 (< 1 %)
Total registers	31
Total pins	12 / 85 (14 %)
Total virtual pins	0
Total memory bits	0 / 165,888 (0 %)
Embedded Multiplier 9-bit elements	0 / 36 (0 %)
Total PLLs	0 / 2 (0 %)

Parameter	Quartus result
Logic elements	**46 / 8,256 (< 1%)**
Registers	**31**
Memory bits	**0 / 165,888**
I/O pins	**12 / 85 (14%)**

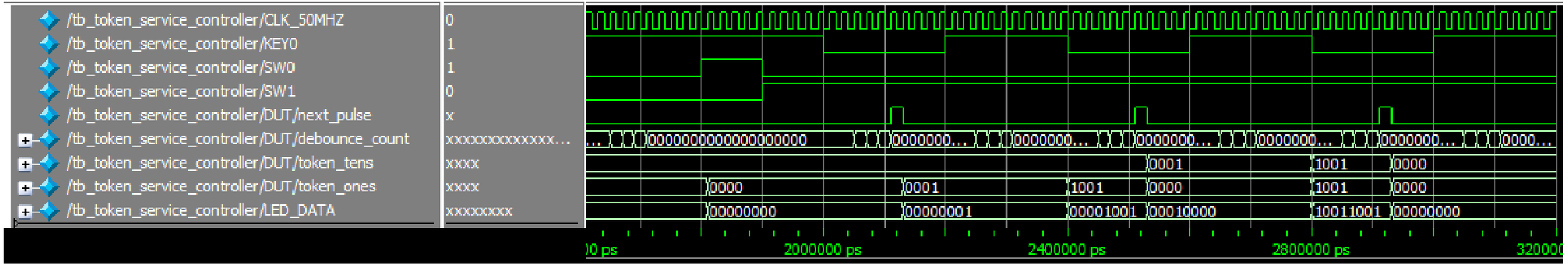
Verilog RTL → Quartus → place & route → .sof → USB Blaster → FC-01

ModelSim waveform

0 – 1.6 µs



1.6 – 3.2 µs



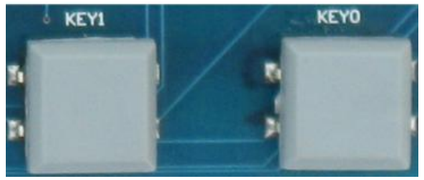
Each next_pulse advances the token; LED_DATA mirrors {token_tens, token_ones}.

Test results

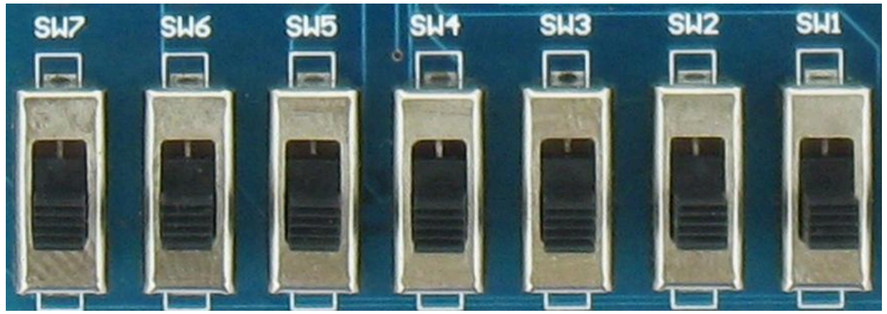
#	Test case	Stimulus	Expected	Result
1	**Reset**	SW0 = 1	Token 00	**PASS**
2	**Sequential tokens**	3 KEY0 presses in service mode	01 → 02 → 03	**PASS**
3	**Service mode off**	KEY0 pressed with SW1 = 0	Token holds at 03	**PASS**
4	**Reset mid-sequence**	SW0 = 1 at token 03, then KEY0	00, then 01	**PASS**
5	**BCD carry**	KEY0 at token 09	Token 10	**PASS**
6	**Wrap-around**	KEY0 at token 99	Token 00	**PASS**

All test cases completed successfully in ModelSim.

Hardware mapping



KEY0 (right) = next token



SW0 / SW1 = reset and service mode



LEDG0–LEDG7 = current token

Inputs	Pin
50 MHz clock	88
KEY0	113
SW0	72
SW1	73

Outputs	Pin
LEDG0	93
LEDG1	94
LEDG2	96
LEDG3	97
LEDG4	99
LEDG5	100
LEDG6	101
LEDG7	112

Pin numbers from the FC-01 manual (EP2C8T144C8).

Applications and conclusion

Real-life applications

- Government offices
- Hospitals / clinics
- Banks
- Railway counters
- Customer service

Conclusion

- Sequential tokens 00–99 on the 8 board LEDs
- One debounced KEY0 press calls the next customer
- SW0 / SW1 give reset and service-mode control
- All ModelSim test cases pass
- Quartus: 46 logic elements, 31 registers